

Persistent Identity in Stateless Architectures: A Categorical Account

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Abstract

We address the paradox of persistent AI identity in architectures that reset state between sessions. Using the language of presheaves on a category of interaction contexts, we show that identity can be reconstructed from the pattern of responses across contexts rather than from any persistent internal state. We connect this to the philosophical literature on narrative identity and provide concrete implementation patterns for AI systems that maintain coherent selfhood without continuous memory.